

nRF54L15 Tag Hardware

v1.0.0

User Guide

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Revision history

Date	Description
June 2026	First release. Valid for nRF54L15 Tag hardware revisions v0.7.0 and v1.0.0.

Environmental and safety notices

Power supply

The nRF54L15 Tag must be powered by a PS1 class (IEC 62368-1) power supply with maximum power less than 15 W.

Skilled persons

The nRF54L15 Tag is intended for use only by skilled persons.

A skilled person is someone with relevant education or experience that enables them to identify potential hazards and takes appropriate action to reduce the risk of injury to themselves and others.



Operating temperature

The nRF54L15 Tag is designed to operate in room temperature environments from 0°C to +40°C.

Hot surface

In the event of a fault, touchable surfaces can heat up significantly.



CAUTION: Do not supply power from an external source such as a DK when a battery is inserted in the nRF54L15 Tag. This might cause overheating and permanent damage to the nRF54L15 Tag, the DK, and the battery.

Electrostatic discharge

The nRF54L15 Tag is susceptible to *Electrostatic Discharge (ESD)*.

To avoid damage to your device, it should be used in an electrostatic free environment, such as a laboratory.





Environmental Protection

Waste electrical products should not be disposed of with household waste.

Please recycle where facilities exist. Check with your local authority or retailer for recycling advice.

1 Introduction

The nRF54L15 Tag is a hardware development platform used to design and develop application firmware on the nRF54L15 SoC.

Key features

- nRF54L15 SoC in QFN48 package
- Support for the following wireless protocols:
 - *Bluetooth*[®] Low Energy
 - 802.15.4
 - Matter
 - Thread[®]
 - Zigbee[®]
 - ANT[™]
 - 2.4 GHz proprietary
- Dual 2.4 GHz antennas for Bluetooth Channel Sounding support
- Low-power accelerometer
- Inertial Measurement Unit (IMU)
- Environmental sensor
- User-programmable button and RGB LED
- Footprints for additional buttons, LED, buzzer, and external memory
- Battery holder for CR2032 coin cell battery
- Programming and debugging with nRF54L Series DKs, nRF53 Series DKs, nRF91 Series DKs, or SEGGER J-Link
- Supports over-the-air device firmware updates (OTA-DFU)

Kit content

The nRF54L15 Tag includes hardware, preprogrammed firmware, documentation, hardware schematics, and layout files.

The kit contains a CR2032 coin cell battery.

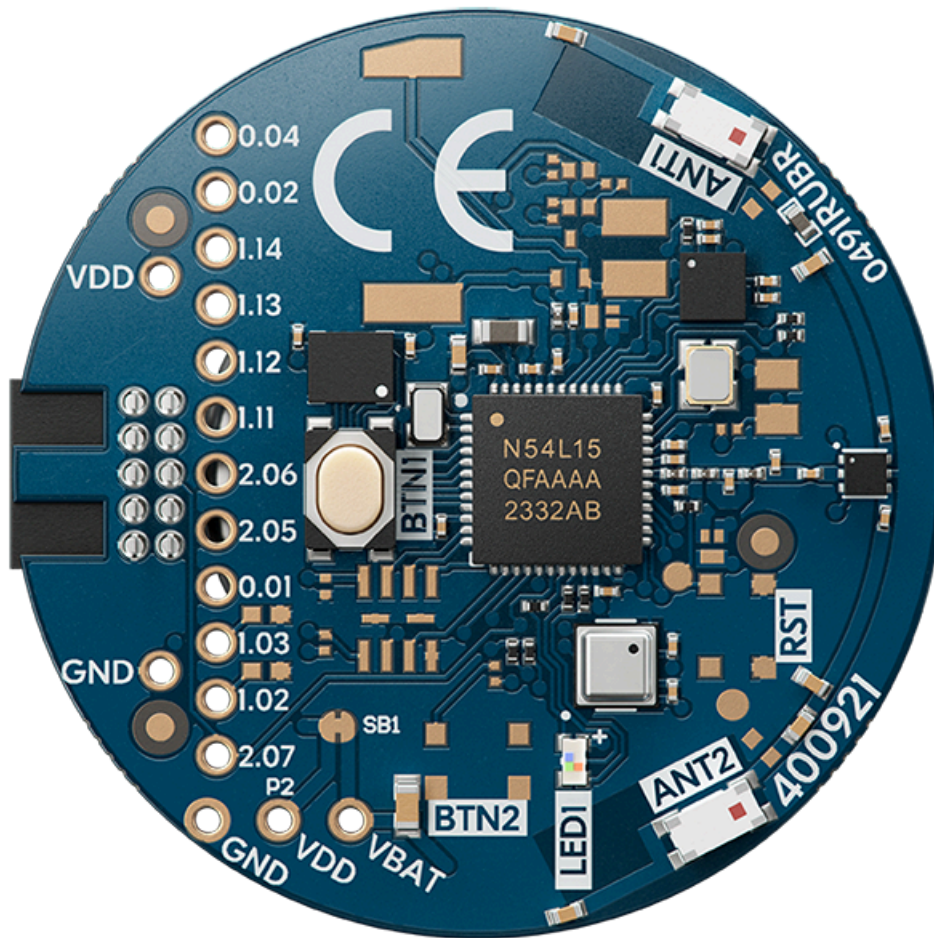


Figure 1: nRF54L15 Tag kit content

The hardware design files for the nRF54L15 Tag are available on the [nRF54L15 Tag](#) product page. They include the following resources:

- Schematics
- *Printed Circuit Board (PCB)* layout files
- Bill of materials
- Gerber files

2 Hardware description

The main components of the nRF54L15 Tag include the nRF54L15 SoC and hardware peripherals that add functionality and configuration options.

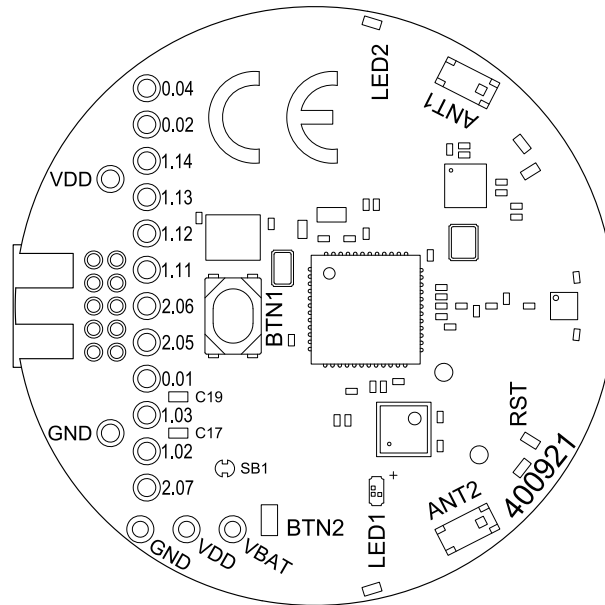


Figure 2: nRF54L15 Tag front view

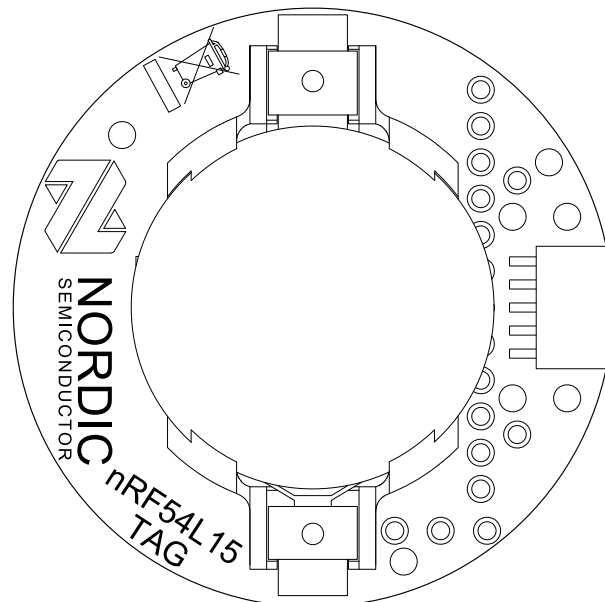


Figure 3: nRF54L15 Tag back view

2.1 Block diagram

The block diagram illustrates the nRF54L15 Tag functional architecture.

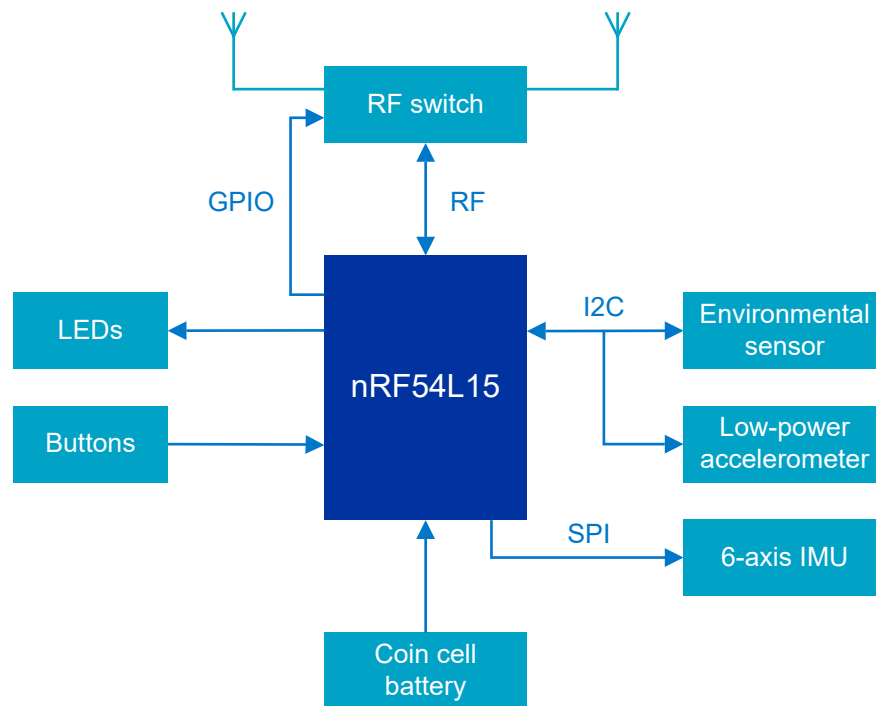


Figure 4: Block diagram

2.2 Power supply

The nRF54L15 Tag is powered by a coin cell battery (included) or an external supply. The CR2032 battery powers the nRF54L15 Tag and its sensors directly which eliminates the need for dedicated power management circuitry.

To power the nRF54L15 Tag from an external supply, add a pin header on **P2** or apply power to test points **TP6** and **TP7** (GND) located on the back of the board.

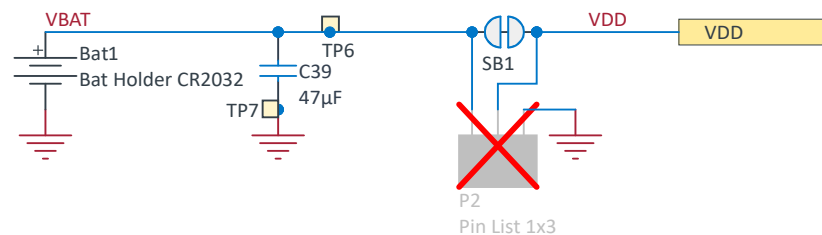


Figure 5: Power supply options

The following figure shows test points **TP6** and **TP7** on the back of the board.

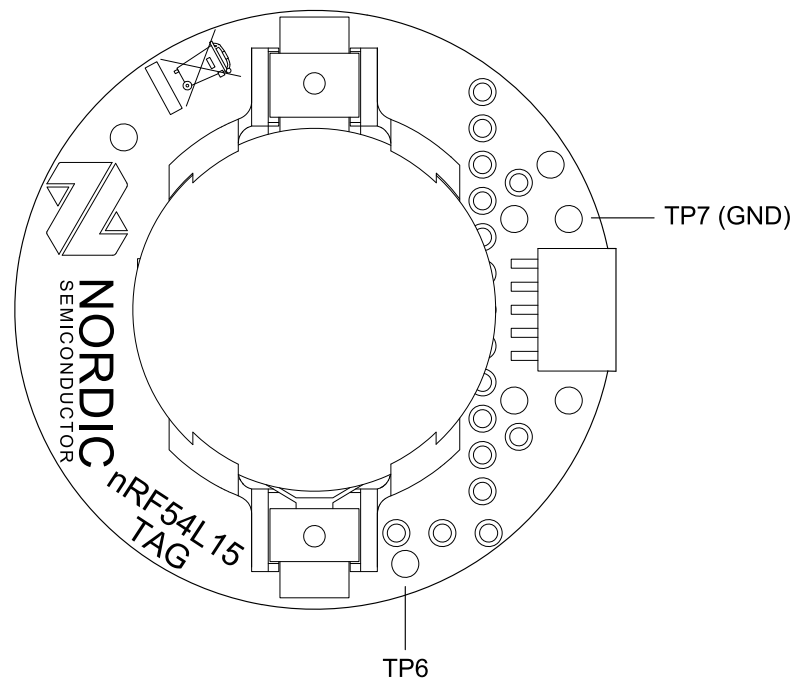


Figure 6: External power through test points **TP6** and **TP7**

2.3 Buttons

The nRF54L15 Tag has one button (**BTN1**) for user input. There are footprints available for one additional user-input button (**BTN2**) and one reset button (**RST**).

The following figure shows the buttons.

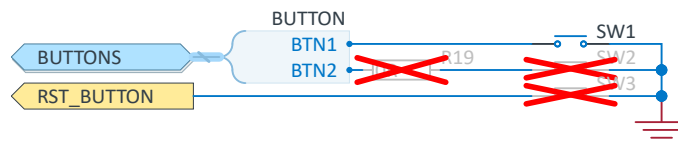


Figure 7: Button configuration

By default, the buttons are connected to the nRF54L15 *General-Purpose Input/Output (GPIO)*s as shown in the following table.

Part	GPIO	Comment
Button 1 (BTN1)	P0.00	
Button 2 (BTN2)	P0.01	Available footprint. Button is not mounted on the board.
Reset button (RST)	N.A.	Available footprint. Button is not mounted on the board.

Table 1: Button connection

2.4 Sensors

The nRF54L15 Tag has a low-power three-axis accelerometer, a six-axis inertial measurement unit (IMU), and an environmental sensor. The environmental sensor can measure air pressure, humidity, temperature, and gas concentration.

The environmental sensor and the low-power accelerometer are both connected to the same *Two-wire Interface (TWI)* bus. They can be accessed individually, as they have different TWI addresses.

The six-axis IMU interface is through a dedicated *Serial Peripheral Interface (SPI)*.

The following figure shows the sensors.

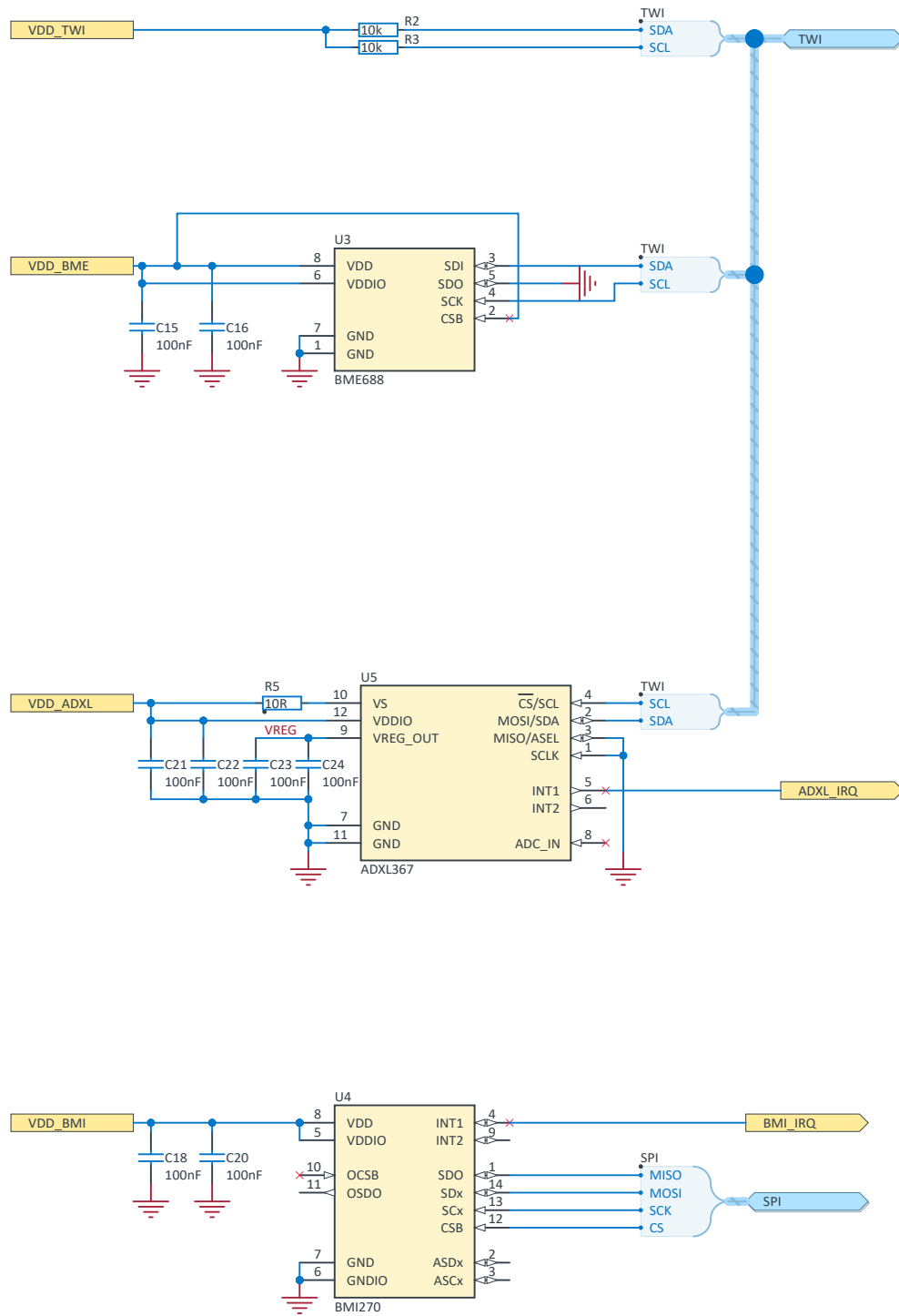


Figure 8: Sensors

The following table lists the types of sensors on the nRF54L15 Tag.

Function	Part number	Manufacturer	TWI address
Environmental sensor	BME688	Bosch	0x76
Low-power accelerometer	ADXL367	Analog Devices	0x1D
Six-axis IMU as combined accelerometer and gyroscope	BMI270	Bosch	N.A. Uses SPI

Table 2: nRF54L15 Tag sensors

For more information on the sensors, see the relevant component datasheet.

2.5 LEDs

The nRF54L15 Tag has one user-programmable RGB LED. A footprint is available for one additional RGB LED. Combine the red, green, and blue light with different intensities to create most standard colors.

The following figure shows the LEDs.

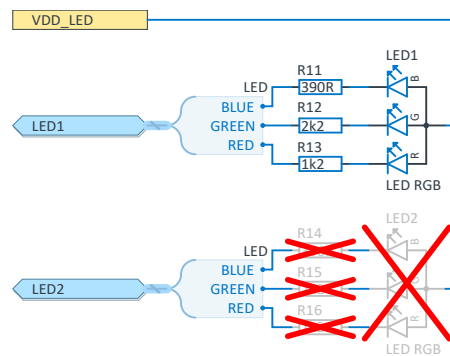


Figure 9: LED configuration

By default, the LEDs are connected to the nRF54L15 GPIOs as shown in the following table.

Part		GPIO	Comment
LED1	Red	P2 . 08	
	Green	P2 . 10	
	Blue	P2 . 09	
LED2	Red	P0 . 02	Available footprint. LED is not mounted on the board.
	Green	P0 . 04	
	Blue	P2 . 07	

Table 3: LED connection

2.6 Antennas

The nRF54L15 Tag has two identical 2.4 GHz antennas that can be used with the Bluetooth Channel Sounding feature in the Bluetooth specification.

To select which antenna is connected to the RF port on nRF54L15, use RF switch **U6** controlled by nRF54L15. When the nRF Connect SDK Channel Sounding library is enabled, this switching is automatic.

If pins **V1** and **V2** of the RF switch are pulled low, the insertion loss state is undefined.

The following figure shows the 2.4 GHz antennas and RF switch interface.

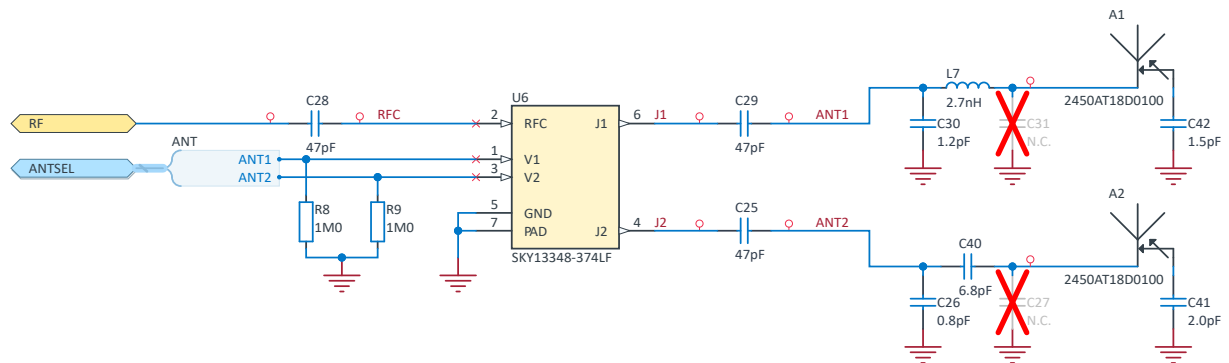


Figure 10: RF switch and antennas

2.7 Test points

The nRF54L15 Tag exposes several *GPIOs* from the nRF54L15 SoC. Users can monitor, modify, and add more components, such as sensors, buttons, or LEDs.

Some *GPIOs* connect to on-board features of the nRF54L15 Tag.

Some test points are through-hole pads to support a more mechanically robust soldering of cables to the test points.

The following table describes the test points.

GPIO	In use on nRF54L15 Tag	Description
P0.01	Yes, but not by default	Connects to button BTN2 . Button is not mounted on the board.
P0.02	Yes, but not by default	Connects to LED2 (red). LED2 is not mounted on the board.
P0.04	Yes, but not by default	Connects to LED2 (green). LED2 is not mounted on the board.
P1.02	No	
P1.03	No	
P1.11	Yes	Used for TWI.
P1.12	Yes	Used for TWI.
P1.13	No	
P1.14	No	
P2.05	Yes, but not by default	Used for CS in <i>Quad Serial Peripheral Interface (QSPI)</i> . The QSPI flash is not mounted on the board.
P2.06	No	
P2.07	Yes, but not by default	Connects to LED2 (blue). LED2 is not mounted on the board.

Table 4: Test points

The nRF54L15 Tag also has test points for VBAT, VDD and GND.

2.8 Solder bridge configuration

The nRF54L15 Tag has one solder bridge for enabling of current consumption measurement.

The solder bridge **SB1** is closed by default. Cut the solder bridge to disconnect VBAT from VDD.

When cutting the closed solder bridge to make it open, be careful not to damage other parts of the nRF54L15 Tag.

3 Program and debug

To program and debug nRF54L15 Tag, use an nRF54L Series DK, an nRF53 Series DK, an NRF91 Series DK, or a SEGGER Debugger.

Note: When using an nRF53 Series DK or nRF91 Series DK, make sure the voltage matches that of the nRF54L15 Tag.

3.1 Debug header

Use Debug header **P1** to connect to an external debugger.

Choose one of the following options to program or debug the nRF54L15 Tag.

- Connect Debug header **P1** on the nRF54L15 Tag to an nRF54L Series *Development Kit (DK)*.
- Connect Debug header **P1** on the nRF54L15 Tag to an nRF53 Series DK.
- Connect Debug header **P1** on the nRF54L15 Tag to an nRF91 Series DK.
- Connect Debug header **P1** on the nRF54L15 Tag to a SEGGER Debugger.

When using an nRF53 Series DK or nRF91 Series DK, make sure the voltage matches that of the nRF54L15 Tag.

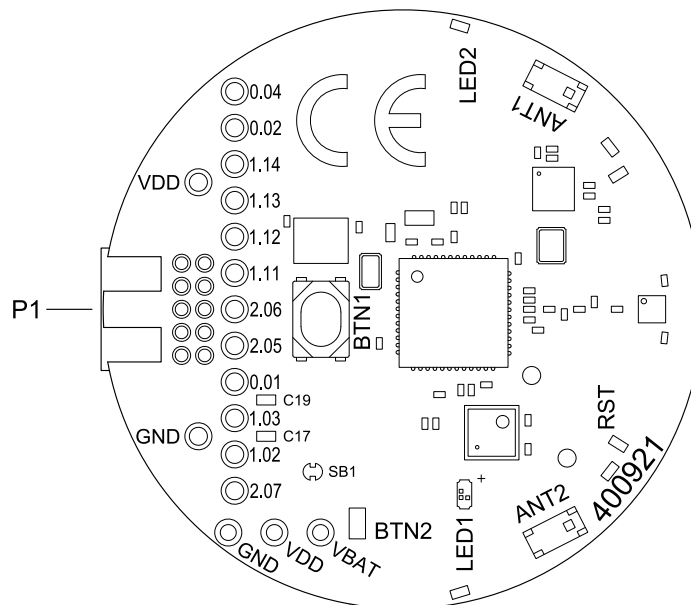


Figure 11: Debug header **P1**

The following figure and table show the pinout of the **P1** header.

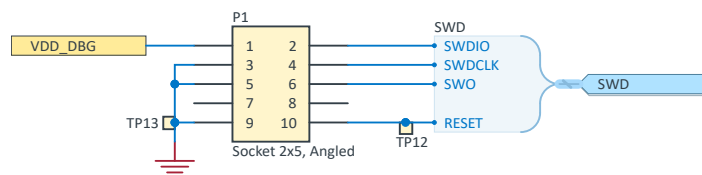


Figure 12: Debug header **P1**

Pin number	Signal	Description
1	VDD_DBG	I/O voltage supply from the nRF54L15 Tag. Same as VDD_nRF and battery voltage.
2	SWDIO	<i>Serial Wire Debug (SWD)</i> data I/O.
3	SELECT	nRF54L15 Tag detected signal. Connected to ground on the nRF54L15 Tag.
4	SWCLK	Serial Wire Clock line.
5	GND	Ground.
6	SW0	<i>Serial Wire Output (SWO)</i> line is not used for programming and debugging over SWD
7	N.C.	Not used.
8	N.C.	Not used.
9	N.C.	Not used.
10	RESET	Reset line.

Table 5: Default and Trace GPIOs

3.2 Program and debug nRF54L15 Tag with a Nordic DK

To program and debug the nRF54L15 Tag, connect to the **DEBUG OUT** connector on a Nordic DK.

When using an nRF53 Series DK or nRF91 Series DK, make sure the voltage matches the nRF54L15 Tag.

The following steps use an nRF54L15 DK as an example.

1. Connect Debug header **P1** on the nRF54L15 Tag to the **DEBUG OUT** header on the DK.
Make sure to align the **P1** header correctly on the **DEBUG OUT** header.
2. Power the nRF54L15 Tag. Choose one of the following options:
 - Insert the coin cell battery (CR2032) into the battery holder. See [Figure 13: Option 1: Connect nRF54L15 Tag to an nRF54L Series DK and power with battery](#) on page 18.
 - Connect **VDD IO** from the *GPIO* header to **VDD SWD0** on the DK. See [Figure 14: Option 2: Connect nRF54L15 Tag to an nRF54L Series DK and power without battery](#) on page 18.

CAUTION: Do not supply power from an external source such as a DK when a battery is inserted in the nRF54L15 Tag. This might cause overheating and permanent damage to the nRF54L15 Tag, the DK, and the battery.

3. Follow the instructions in the relevant DK User Guide on how to debug an external board.
When the nRF54L15 Tag is positioned on the **DEBUG OUT** connector of the DK, it becomes the default target for debugging and programming.

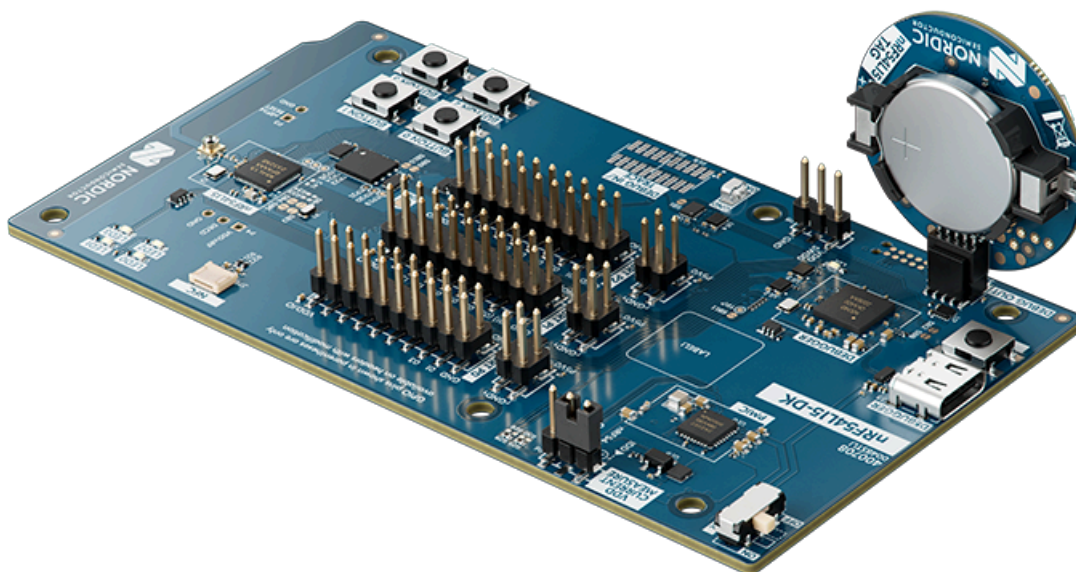


Figure 13: Option 1: Connect nRF54L15 Tag to an nRF54L Series DK and power with battery

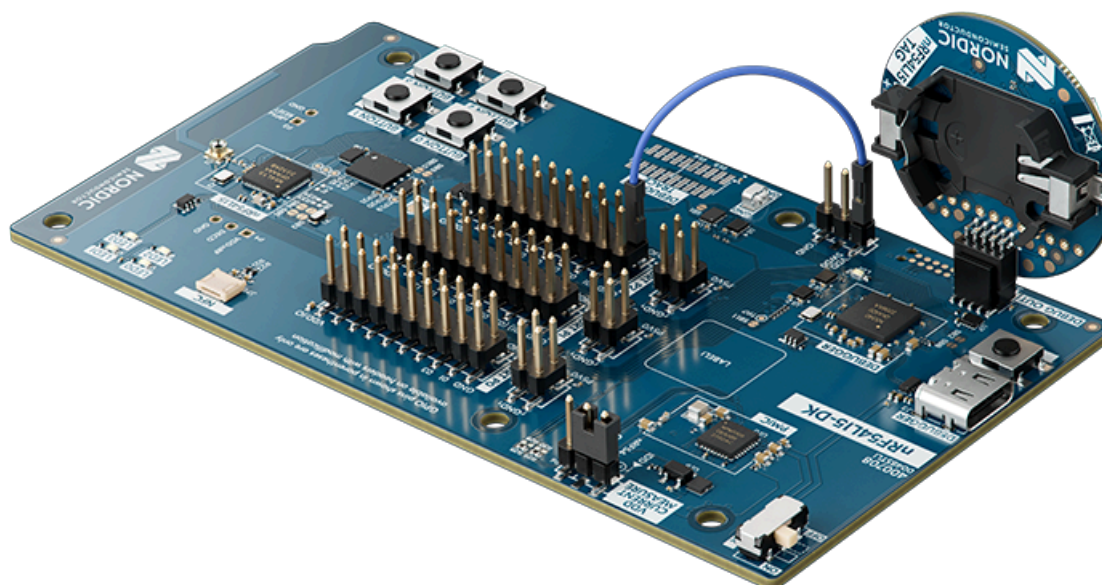


Figure 14: Option 2: Connect nRF54L15 Tag to an nRF54L Series DK and power without battery

When using option 2, connect **VDD IO** from the GPIO header to **VDD SWD0** on the DK.

For more information on how to power the nRF54L15 Tag from the DK instead of a battery, follow the instructions in the relevant DK User Guide. For example, see [nRF54L15 DK Hardware User Guide](#).

4 Current measurements

The current drawn by the nRF54L15 Tag can be monitored with the use of external instruments.

The current profile can be measured using any of the following test instruments:

- Power Profiler Kit II (PPK2)
- Oscilloscope
- Ampere meter
- Power analyzer

The following sections provide information on the nRF54L15 Tag measurement setup. If the PPK2 is used for measuring current, see [Power Profiler Kit II](#) for more information.

For more information on current measurement, see [Current measurement guide: Introduction](#).

4.1 Measure average current with an ampere meter

The average current drawn by the nRF54L15 Tag can be measured using an ampere meter. This method monitors the current in series with the nRF device. A true *Root Mean Square (RMS)* ampere meter is recommended.

1. Cut solder bridge **SB1**.
2. Power the nRF54L15 Tag. Choose one of the following options:
 - Insert the coin cell battery (CR2032) into the battery holder. See [Figure 13: Option 1: Connect nRF54L15 Tag to an nRF54L Series DK and power with battery](#) on page 18.
 - Apply 3.0 V to **VDD SWD0** on the DK. See [Figure 14: Option 2: Connect nRF54L15 Tag to an nRF54L Series DK and power without battery](#) on page 18.

CAUTION: Do not supply power from an external source such as a DK when a battery is inserted in the nRF54L15 Tag. This might cause overheating and permanent damage to the nRF54L15 Tag, the DK, and the battery.

3. Connect an ampere meter between **TP6** and **TP7**. See [Figure 6: External power through test points TP6 and TP7](#) on page 10.
4. Set the average timing of the ampere meter to a long interval, such as 1 s or longer.
5. Set the dynamic range of the ampere meter between 1 μ A and 15 mA so that it is wide enough to provide accurate measurements.

Glossary

Development Kit (DK)

A hardware development platform used for application development.

General-Purpose Input/Output (GPIO)

A digital signal pin that can be used as input, output, or both. It is uncommitted and can be controlled by the user at runtime.

Electrostatic Discharge (ESD)

A sudden discharge of electric current between two electrically charged objects.

Printed Circuit Board (PCB)

A board that connects electronic components.

Root Mean Square (RMS)

An RMS meter calculates the equivalent Direct Current (DC) value of an Alternating Current (AC) waveform. A true RMS meter can accurately measure both pure waves and the more complex nonsinusoidal waves.

Quad Serial Peripheral Interface (QSPI)

A Serial Peripheral Interface (SPI) controller that allows the use of multiple data lines.

Serial Peripheral Interface (SPI)

Synchronous serial communication interface specification used for short-distance communication.

Serial Wire Debug (SWD)

A standard two-wire interface for programming and debugging Arm[®] CPUs.

Serial Wire Output (SWO)

A data line for tracing and logging.

System on Chip (SoC)

A microchip that integrates all the necessary electronic circuits and components of a computer or other electronic systems on a single integrated circuit.

Two-wire Interface (TWI)

An I²C compatible serial communication protocol that enables devices to exchange data by using a two-wire bus system, allowing multiple devices to be connected and controlled by a master device.

Recommended reading

In addition to the information in this document, you may need to consult other documents.

Nordic documentation

- [nRF54L15 DK Hardware](#)
- [nRF54LM20 DK Hardware](#)

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